CSE374: Machine Learning Classification Project

Dataset Menu:
Choose one of the following datasets (Datasets details are at the end of the file)

Level 1:

- Banknote Authentication
- Blood Transfusion Service Center
- Rice (Cammeo and Osmancik)
- Climate Model Simulation Crashes

Level 2 (+1 bonus):

- Student Performance
- QSAR Fish Toxicity
- QSAR Biodegradation
- Steel Plates Faults
- Dry Bean Dataset
- Heart Failure Clinical Records
- Shill Bidding Dataset
- Contraceptive Method Choice

Level 3 (+2 bonus):

- Credit Card Default (Taiwan)
- Gas Sensor Array Drift
- Electrical Grid Stability
- Thoracic Surgery Data
- HCV Dataset
- Mushroom Dataset (feature reduction required)
- Steel Industry Energy Consumption (engineered target)
- Landslide Susceptibility Dataset
- Parkinsons Telemonitoring (engineered target)

You may propose another dataset with instructor approval.

Number of students per team: 3
Required project tasks:

Students must produce an end-to-end supervised classification study on one dataset.

1. Problem Framing & Data Understanding

- What is the problem being solved?
- What is the target variable?
- Is it binary or multiclass classification?
- What challenges do you expect (e.g., imbalance, noise)?

2. Data Acquisition & Documentation

- Dataset source (link)
- Any modifications made (e.g., cleaning, label changes, training/test splits)

4. Data Cleaning & Preprocessing

- Handling missing values
- Encoding categorical variables
- Scaling numerical features (if needed)
- Handling class imbalance (if needed)

5. Exploratory Data Analysis (EDA)

Include ONLY 3–5 meaningful plots:
- Class distribution
- 1–2 feature vs target plots
- Correlation heatmap (optional)

Each plot MUST include 2–3 lines of interpretation.

6. Modeling

Implement at least three classifiers of the following:

- Logistic Regression
- Decision Tree
- (SVM, kNN, or Naive Bayes)
-Shallow Neural networks

Use train/validation or cross-validation correctly; avoid test leakage.

7. Performance Evaluation

- Accuracy
- F1-score
- Confusion matrix
- ROC/AUC

8. Model Comparison

Present results in a table and explain which model performed best and why.

9. Error Analysis

Briefly discuss where the model fails and possible reasons.

Inspect misclassified examples (where possible) and characterize common error patterns.

Discuss limitations: data size, noise, bias.

10. AI Tool Usage

Include 2–3 examples of how AI tools were used and what you learned as follows:

1. Use at least one AI tool (e.g., ChatGPT-like assistant, Copilot) as a coding or ideation assistant.
2. Keep a log of prompts and outputs, and write a short reflection:
    - What was helpful?
    - What was wrong or misleading?
    - How did you verify and correct AI-generated code?

11. Deliverables

- Clean notebook (runs end-to-end)
- Report (10 pages)
- Slides
- Contribution table, each team member contribution

Roles and individual responsibilities

Each group of three assigns and documents roles. Students can help each other, but each role must have clearly attributable work.

1. **Role 1 – Data & Preprocessing Lead**
    - **Lead** data loading, cleaning, and preprocessing pipeline.
    - Design train/validation/test split strategy and any cross-validation.
    - Document all preprocessing steps in the report and in a separate "data_preprocessing" section of the notebook.
    - Individual deliverable: 1–2 page "Data & Preprocessing Note" summarising decisions and justifying design choices.
2. **Role 2 – EDA & Visualization Lead**
    - Design and implement the key descriptive statistics and visualizations.
    - **Lead** the narrative around data understanding and feature–label relationships.
    - Ensure plots are labelled, readable, and directly support conclusions.
    - Individual deliverable: "EDA & Insights Note" with 3–5 figures and commentary.
3. **Role 3 – Modeling & Evaluation Lead**
    - Implement and tune multiple classification models.
    - Coordinate comparison tables, confusion matrices, ROC curves, and feature importance.
    - **Lead** the discussion of model performance, limitations, and potential improvements.
    - Individual deliverable: "Modeling & Evaluation Note" summarising model design and results.
4. **Shared group responsibilities**
    - All three students must contribute to:
        - The final integrated notebook(s).
        - The written report (sections clearly attributed).
        - The AI-tool reflection section (each student contributes at least one example).
        - The final presentation (each member presents part of the work).

Students should submit a short "contribution statement" with the report indicating who did what.

AI tools: allowed and expected use

- AI tools **are allowed and encouraged as assistants**, not as a source of final answers. Acceptable uses include:
  - Brainstorming preprocessing ideas, model choices, or visualization strategies.
  - Drafting boilerplate code (e.g., training loops, plotting templates).
  - Getting explanations of error messages or library functions.
- Unacceptable uses include:
  - Asking AI to do the entire project or to generate the full notebook/report.
  - Submitting AI-generated text or code without understanding or significant modification.

Project Milestones:

1. First Deliverable:
   A small report and Notebook for steps from 1-5 (till EDA) and implement one classifier only

2. Final Deliverable
   Complete all requirements and deliverables

Grading rubric

Criterion	Excellent (A range)	Good (B range)	Satisfactory (C range)	Poor (D/F)	Weight
Problem framing & dataset understanding	Clear, accurate description of domain, features, labels, and challenges	Mostly clear; minor gaps or vague aspects	Basic description; several important aspects missing	Incomplete or incorrect description	10%
Data cleaning & preprocessing	Appropriate handling of missing values, encoding, scaling; decisions well-justified	Mostly appropriate; some ad-hoc steps or limited justification	Minimal cleaning; questionable choices; little justification	Major issues or no clear preprocessing	15%

EDA & visualization	Insightful plots and stats tightly linked to narrative and decisions	Reasonable plots; some connection to modeling decisions	Basic plots, little interpretation	Little or no EDA or unreadable plots	10%
Modeling breadth & correctness	≥3 classifier families implemented correctly; sound train/val/test design	2–3 models, minor implementation or validation issues	Only 1–2 simple models; issues with splits or misuse of test data	Incorrect model usage, data leakage, or non-reproducible pipeline	20%
Evaluation & comparison	Appropriate metrics, confusion matrices, ROC (if binary); results compared and analysed	Metrics reported, some comparison but limited insight	Minimal metrics; weak or purely numeric comparison	Little or no evaluation or misinterpreted metrics	15%
Use of AI tools & reflection	AI used meaningfully; examples documented; critical reflection on benefits/risks	AI use documented but reflection is superficial	AI mentioned but poorly documented; unclear what was actually used	No AI use documented or clear over-reliance with no reflection	5%
Code quality & reproducibility	Clean, well-commented, modular, runs end-to-end from raw data to results	Mostly clean; minor reproducibility issues	Runs with small fixes; messy structure	Does not run or major missing pieces	5%
Report quality	Clear structure, good writing, correct figures/tables	Generally clear; minor writing or organization issues	Hard to follow, poorly structured; some missing sections	Very poor writing or missing key sections; weak or no presentation	5%
Presentation	Clear and engaging explanation of problem, data, models, and	Mostly clear explanation; most visuals and results understandable;	Some clarity issues; key results or model explanations	Confusing or incomplete explanation; visuals missing or misleading; poor delivery; unable	15%

	results; visuals enhance understanding; confident delivery; handles questions effectively	delivery mostly confident; answers questions adequately.	incomplete; visuals weak; delivery hesitant; struggles with questions.	to answer questions.	

Datasets

1. **Blood Transfusion Service Center (donor conversion)** – UCI / PMLB
    - Link: https://www.openml.org/search?type=data&sort=runs&id=1464 or via PMLB id blood-transfusion-service-center.[arxiv]
    - Size: 748 instances, 4 features.
    - Features: RFM-style donation history (recency, frequency, monetary, time).
    - Target: Whether a donor donated in the next campaign (binary).
2. **Steel Plates Faults** – UCI / OpenML
    - Link: https://archive.ics.uci.edu/dataset/222/steel+plates+faults or OpenML id 1463.[arxiv]
    - Size: 1941 instances, 27 numeric features.
    - Features: Geometric characteristics of steel plates.
    - Target: 7 fault types (multi-class).
3. **QSAR Fish Toxicity**
    - Link: https://archive.ics.uci.edu/dataset/278/qsar+fish+toxicity.[arxiv]
    - Size: 908 instances, 6 molecular descriptors.
    - Features: Physicochemical properties.
    - Target: Toxicity level (ordinal classification).
4. **QSAR Biodegradation**
    - Link: https://archive.ics.uci.edu/dataset/155/qsar+biodegradation.[arxiv]
    - Size: 1055 instances, 41 descriptors.
    - Features: Topological and molecular descriptors.
    - Target: Ready biodegradability (binary).
5. **Climate Model Simulation Crashes**
    - Link: https://archive.ics.uci.edu/dataset/294/climate+model+simulation+crashes.[arxiv]
    - Size: 540 instances, 18 features.
    - Features: Climate model input parameters.
    - Target: Simulation crashed vs ran successfully (binary).

6. **Hepatitis C Virus (HCV) Data**
    - Link: https://archive.ics.uci.edu/dataset/515/hcv+data.[arxiv]
    - Size: ~615 patients, 13 lab measurements + demographics.
    - Features: Blood test values, age, etc.
    - Target: Disease category (6 classes: healthy, fibrosis stages, cirrhosis).
7. **Heart Failure Clinical Records**
    - Link: https://archive.ics.uci.edu/dataset/519/heart+failure+clinical+records.[arxiv]
    - Size: 299 patients, 12 features.
    - Features: Age, ejection fraction, creatinine, etc.
    - Target: Death event (binary).
8. **Parkinsons Telemonitoring (binary version you define)**
    - Link: https://archive.ics.uci.edu/dataset/190/parkinsons+telemonitoring.[arxiv]
    - Size: ~5800 rows, 26 biomedical voice measures.
    - Features: Voice signal measures over time.
    - Suggested target: Create your own label such as "high vs low symptom severity" based on UPDRS thresholds (explain your design).
9. **Banknote Authentication**
    - Link: https://archive.ics.uci.edu/dataset/267/banknote+authentication.[arxiv]
    - Size: 1372 instances, 4 numerical wavelet features.
    - Features: Image statistics of banknotes.
    - Target: Genuine vs forged (binary).
10. **Gas Sensor Array Drift at Different Concentrations**
    - Link: https://archive.ics.uci.edu/dataset/295/gas+sensor+array+drift+dataset.[arxiv]
    - Size: 13,910 samples, 128 features.
    - Features: Sensor responses over time.
    - Target: Gas type (6 classes).
11. **Electrical Grid Stability Simulated Data**
    - Link: https://archive.ics.uci.edu/dataset/374/electrical+grid+stability+simulated+data.[arxiv]
    - Size: 10,000 instances, 12 features.
    - Features: Grid load, power, etc.
    - Target: Stable vs unstable (binary).
12. **Dry Bean Dataset**
    - Link: https://archive.ics.uci.edu/dataset/602/dry+bean+dataset.[arxiv]
    - Size: 13,611 instances, 16 features.
    - Features: Shape/texture of bean images summarized as tabular features.
    - Target: 7 bean varieties (multi-class).
13. **Shill Bidding in Online Auctions**
    - Link: https://archive.ics.uci.edu/dataset/579/shill+bidding+dataset.[arxiv]

- Size: ~6321 bids, 13 features.
- Features: Bid patterns, price changes, time gaps.
- Target: Shill vs normal bid (binary).

14. **Rice (Cammeo and Osmancik)**
    - Link: https://archive.ics.uci.edu/dataset/545/rice+cammeo+and+osmancik.[arxiv]
    - Size: 3810 grains, 7 features.
    - Features: Shape metrics of rice grains.
    - Target: Rice variety (binary).
15. **Steel Industry Energy Consumption** (turn into classification by thresholding)
    - Link: https://archive.ics.uci.edu/dataset/851/steel+industry+energy+consumption.[arxiv]
    - Size: ~3500 samples, 29 features.
    - Features: Process variables and energy use.
    - Suggested target: Define "high vs low" energy consumption based on an engineered threshold; justify your decision.
16. **Mushroom Edibility (but filtered / modified)**
    - Link: https://archive.ics.uci.edu/dataset/73/mushroom.[arxiv]
    - Size: 8124 instances, 22 categorical features.
    - Features: Cap color, odor, gill size, etc.
    - Target: Edible vs poisonous (binary).
    - Instructor suggestion: Require students to drop a few "too obvious" features like odor and justify the feature-reduction rationale to make it less trivial.
17. **Landslide Susceptibility**
    - Example: Data.World "Landslide susceptibility mapping in India" (searchable, CSV).[asmedigitalcollection.asme]
    - Size: a few thousand grid cells, 10–20 geo-environmental features (slope, lithology, rainfall).
    - Target: Landslide occurrence yes/no.
18. **Credit Card Clients Default (Taiwan)**
    - Link: https://archive.ics.uci.edu/dataset/350/default+of+credit+card+clients.[arxiv]
    - Size: 30,000 clients, 23 features.
    - Features: Demographics, bill statements, payment status.
    - Target: Default next month (binary).
19. **Student Performance (Secondary Education)**
    - Link: https://archive.ics.uci.edu/dataset/320/student+performance.[arxiv]
    - Size: ~649 students, 30 features.
    - Features: Demographics, study time, absences, grades.
    - Suggested target: Pass/fail or high/low performance based on final grade threshold.
20. **Contraceptive Method Choice**

- Link: https://archive.ics.uci.edu/dataset/30/contraceptive+method+choice.[arxiv]
- Size: 1473 women, 9 features.
- Features: Demographic and family planning history.
- Target: No use, long-term, or short-term contraception (3 classes).

21. **Thoracic Surgery Data**
    - Link: https://archive.ics.uci.edu/dataset/17/thoracic+surgery+data.[arxiv]
    - Size: 470 patients, 17 features.
    - Features: Pre-operative risk factors.
    - Target: 1-year survival after surgery (binary).